regression equation is used to estimate the value of Y corresponding to the known value of X. The line describing this tendency to regress or going back was called by Galton a "**Regression Line**".

Difference between Correlation and Regression

S.No	Correlation	Regression
1	Correlation is the relationship between two or more variables, which vary with the other in the same or the opposite direction	Regression means going back and it is a mathematical measure showing the average relationship between two variables
2	Both the variables X and Y are random variables	Both the variables may be random variables
3	It finds out the degree of relationship between two variables and not the cause and effect relationship.	It indicates the cause and effect relationship between the variables and establishes functional relationship.
4	It is used for testing and verifying the relation between two variables and gives limited information	Besides verification it is used for the prediction of one value, in relation to the other given value.
5	The coefficient of correlation is a relative measure. The range of relationship lies between –1 and +1	Regression coefficient is also relative measure. If we know the value of the independent variable, we can find the value of the dependent variable
6	There may be spurious correlation between two variables.	In regression there is no such spurious regression
7	It has limited application, because it is confined only to linear relationship between the variables	It has wider application, as it studies linear and nonlinear relationship between the variables
8	It is not very useful for further mathematical treatment.	It is widely used for further mathematical treatment

Two Regression lines

$$X \text{ on } Y \Rightarrow X = a + by$$

$$Y \text{ on } X \Rightarrow Y = a + bx$$

Regression line is the line which gives the best estimate of one variable from the value of any other given variable. The line gives the average relationship between the two variables in mathematical form. The line of regression is the line which gives the best estimate to the value of one variable for any specific value of the other variable.

To fit Regression equations X on Y and Y on X the following examples are given

Ex 1: Fit two regression equation

X on Y and Y on X for the following data.

$\bar{X}$=12, $\bar{Y}$=10, σy= 0.2, σx =0.1 and r = 0.85

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